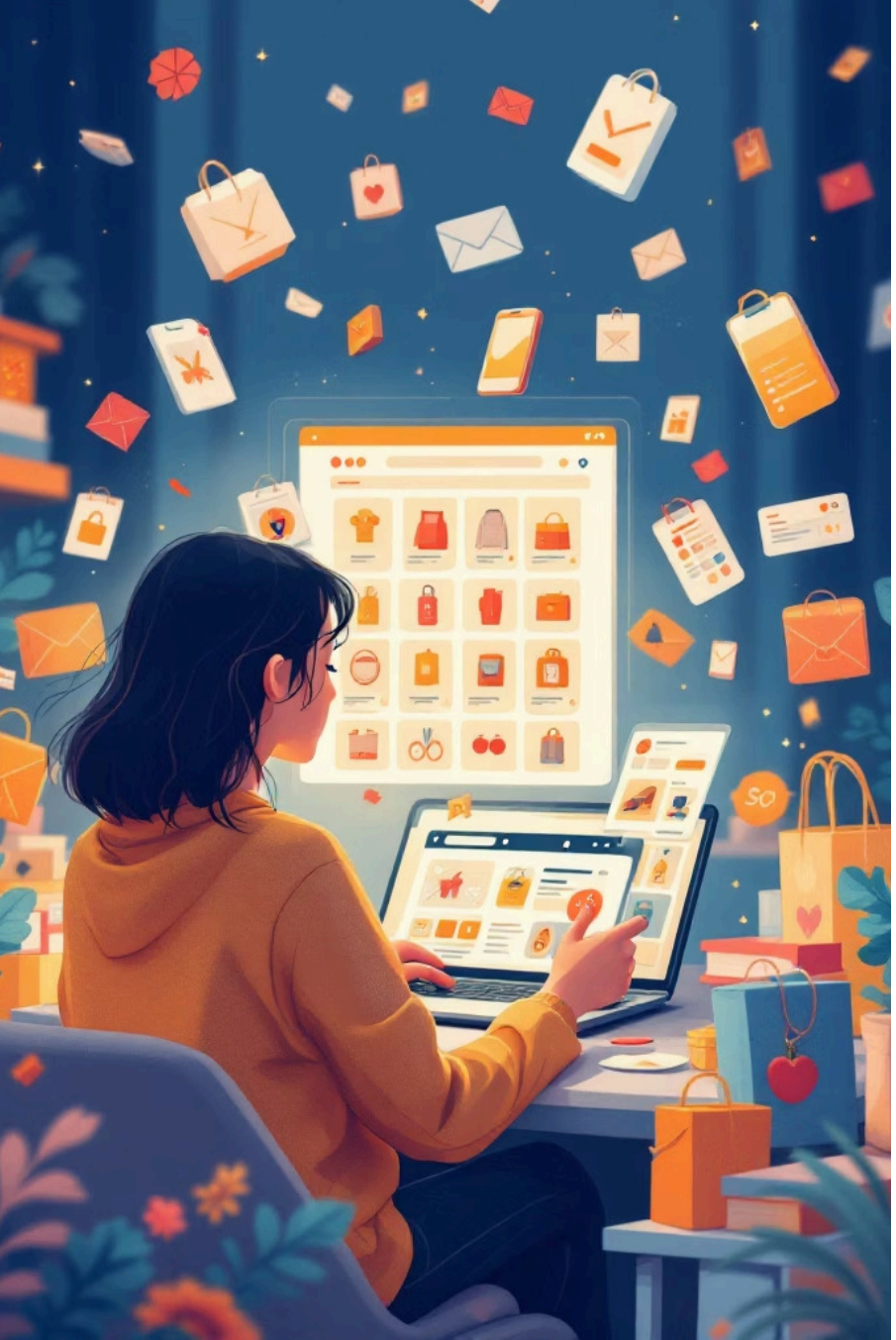

AI-Powered Product Recommendation System

Name: Sanket Patil **Role:** AIML Intern **Context:** Technical project presentation

- Information overload in e-commerce makes discovery slow, repetitive, and fragmented.
- Personalisation improves user experience, relevance, and revenue per session.
- **Problem statement:** recommend the right item to the right user at the right time.

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Case Study Context & Dataset

- E-commerce recommendation problem modelled using a MovieLens-style proxy dataset.
- Scale: approximately 943 users, 1,682 items, and 100K interactions.
- Primary challenges: sparse interaction matrix, scalability, and diverse preference patterns.

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Stakeholder Analysis

Stakeholder	Objective	Expected outcome
Users	Find relevant items quickly	Higher satisfaction and lower search effort
Business	Increase conversion and basket value	Improved revenue and retention
Data Science Team	Build accurate ranking models	Reliable offline and online performance
Product Team	Embed recommendations into journeys	Better engagement across key surfaces

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Recommendation System Paradigms

Approach	Strength	Limitation	Fit
Content-Based	Uses item attributes	Weak novelty, metadata dependence	Good for cold-start items
Collaborative	Learns from interaction patterns	Suffers from sparsity	Best core signal for this project
Hybrid	Combines multiple signals	Higher complexity	Strong future extension

- **Selected:** collaborative filtering, because the behavioural signal is strongest.
- It supports latent preference learning without requiring rich item metadata.

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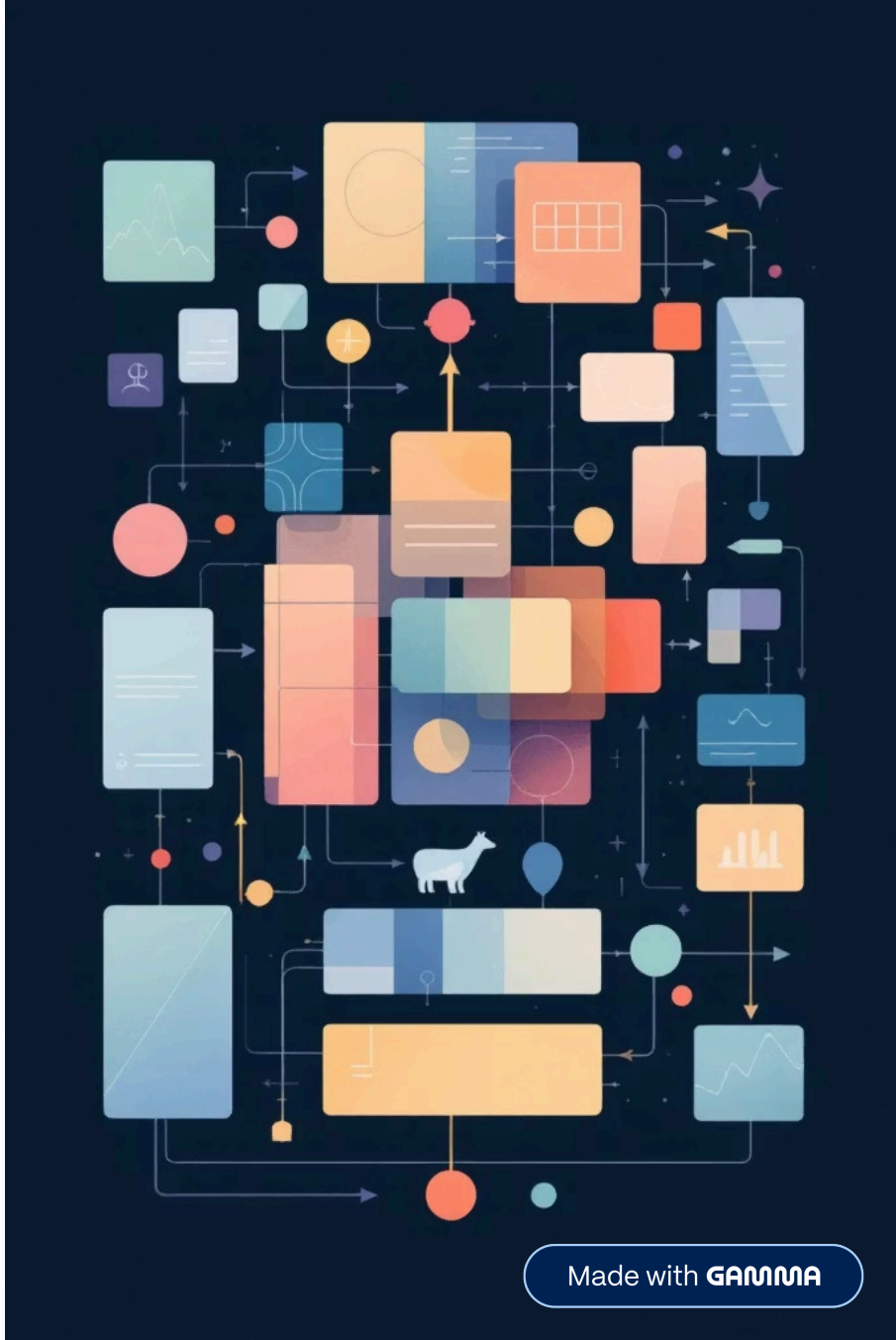
AI Solution Approach: Matrix Factorisation

- Represent user-item interactions as a sparse interaction matrix R.
- Decompose preferences into latent factors: $R \approx U \times V^T$.
- Learn hidden taste dimensions that generalise beyond explicit ratings.
- Use the learned embedding space to rank unseen items for each user.



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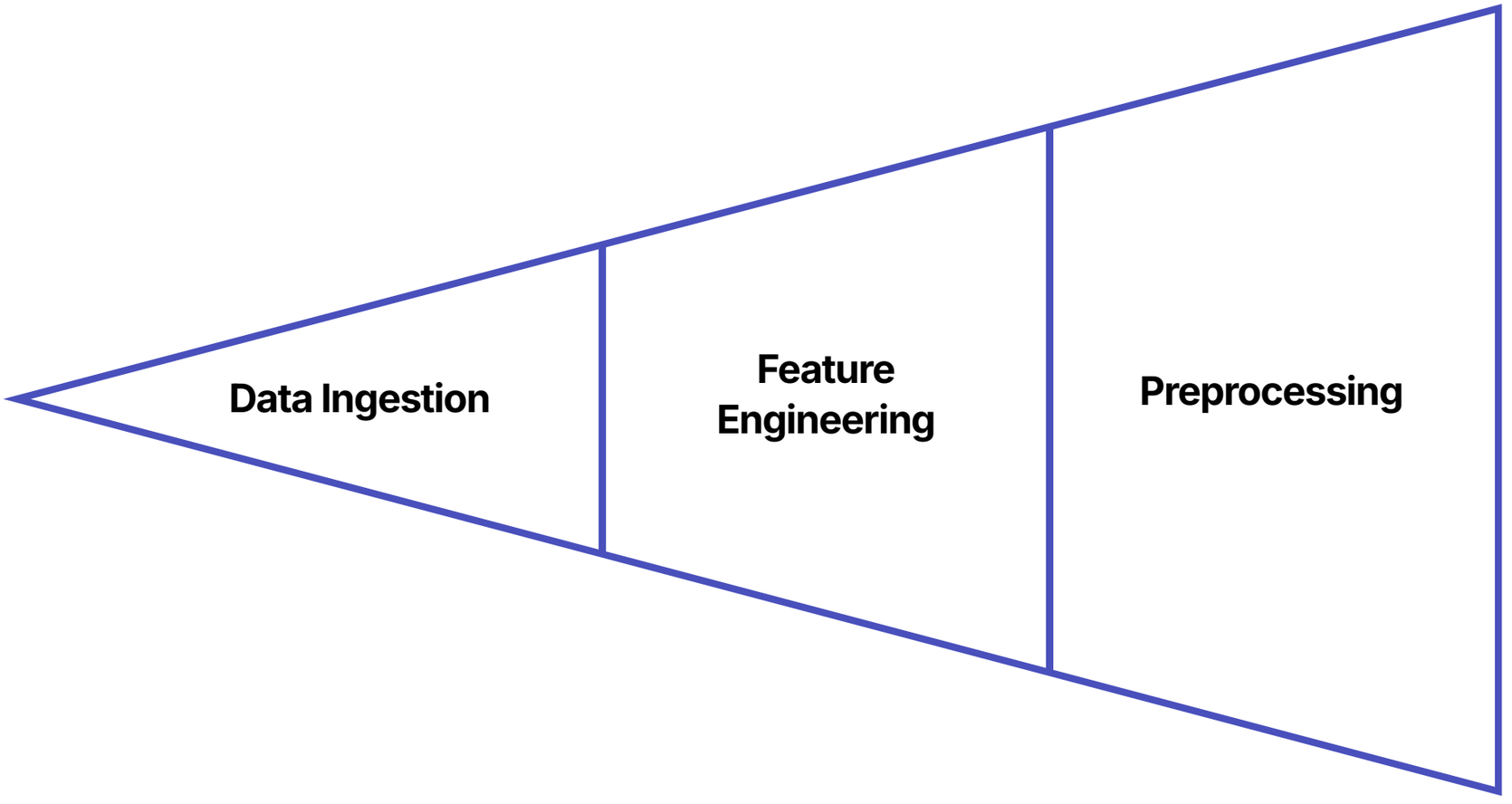
System Architecture



The pipeline converts raw interaction logs into ranked recommendations. Each stage reduces noise, improves signal quality, and exposes the model through a simple product interface.

- **Data:** ratings and interactions.
- **Preprocessing:** cleaning, encoding, deduplication.
- **Model:** matrix factorisation for scoring.
- **Serving:** top-K output exposed in a UI.

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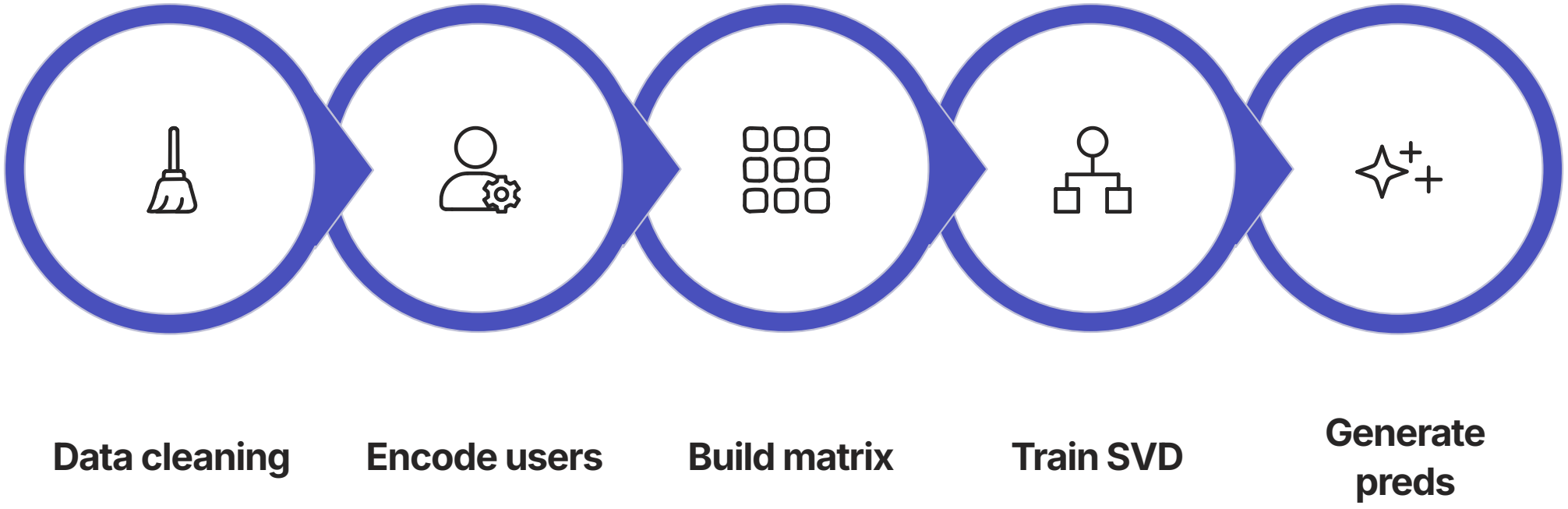
Implementation Workflow



The implementation is intentionally modular: the same pipeline supports offline evaluation, batch scoring, and product-facing recommendations. Filtering previously seen items ensures novelty and avoids duplicate suggestions.

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Demo & Application Interface

- Streamlit UI accepts a **User ID** and returns ranked recommendations instantly.
- Each result includes item title and predicted relevance score.
- Simple interface keeps the model interpretable for non-technical stakeholders.
- Best suited for demo environments, evaluation, and rapid iteration.

Visual: application screenshot placeholder with input panel and ranked output list.

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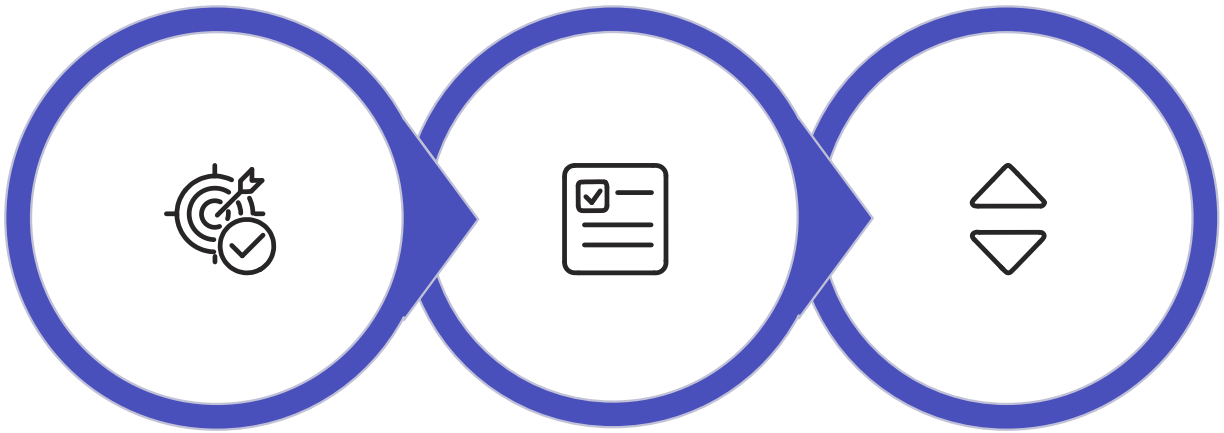
Model Evaluation Metrics



Precision@K **Recall@K** **NDCG@K**

Ranking metrics matter because recommendation quality is not only about predicting relevant items, but ordering them well at the top of the list.

- **Precision@K:** proportion of recommended items that are relevant.
- **Recall@K:** proportion of relevant items recovered within the top-K.
- **NDCG@K:** rewards relevance while accounting for rank position.

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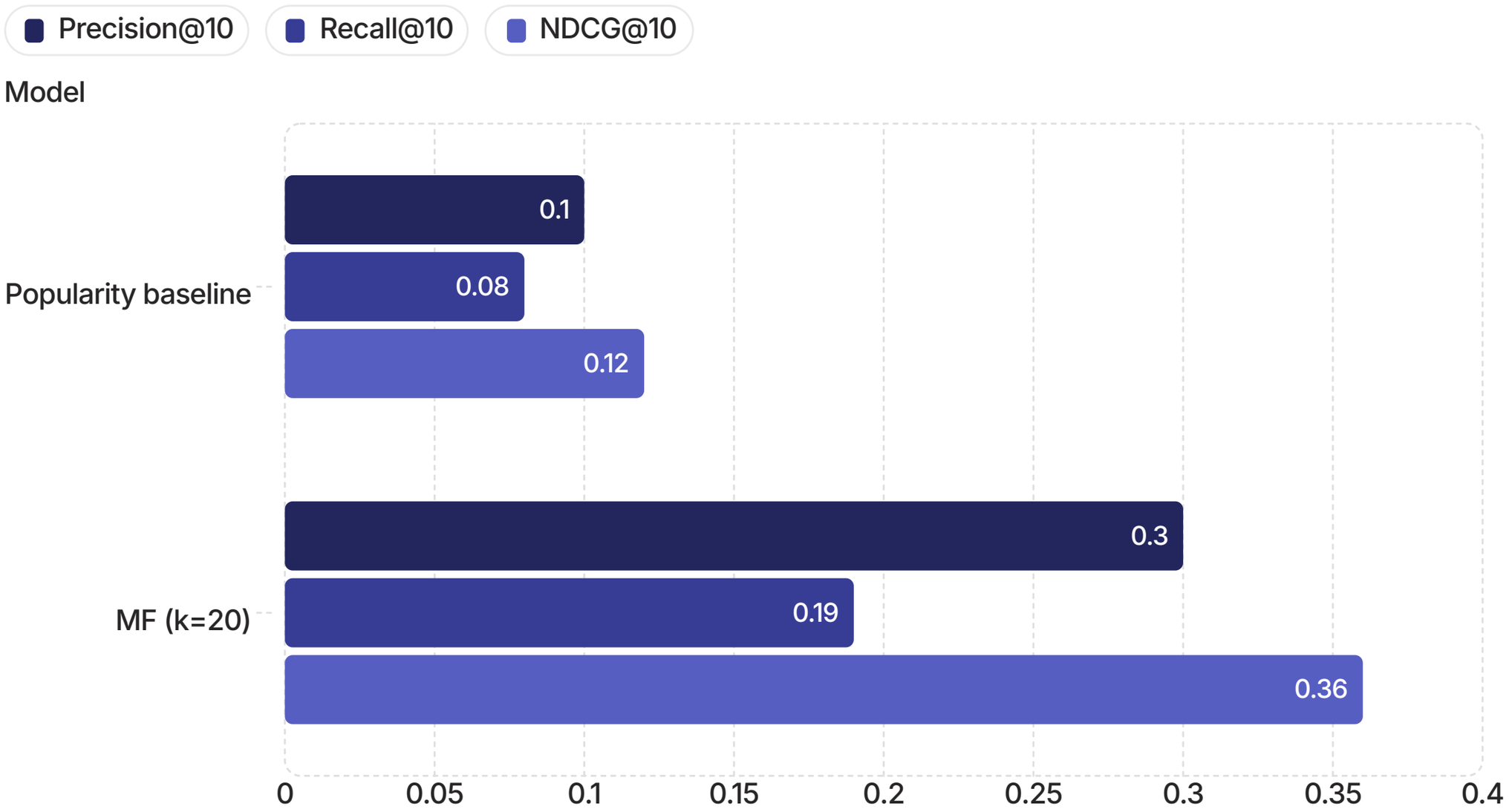
Results & Performance Comparison



Observed result: the matrix factorisation model delivers roughly a 3x uplift over the popularity baseline on precision, with stronger ranking quality across recall and NDCG.

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